

Design and Development of a mHealth Dementia Screening Tool in Resource-Limited Settings

Lessons Learned from Designing for and with Community Health Workers in Peru

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Abstract

Dementia impacts millions worldwide, with rural regions of Latin America facing notable challenges in diagnosis and care. Digital healthcare tools, while potentially transformative, often fail in these areas due to lack of cultural integration, usability issues, and missing features, resulting in ineffective top-down solutions. This case study explores how Community Engagement and Involvement (CEI) principles were applied in the design of a mobile health dementia screening system tailored for rural Peru. For 18 months, we engaged in an iterative participatory process with local community health workers (CHWs) and healthcare professionals. We detail how this process directly informed our design outcomes, ensuring

the solution was adapted to the specific needs of the communities. In addition, the paper highlights key learnings for HCI practitioners and researchers working on similar projects. These insights underscore how participatory design approaches can lead to more effective socio-technical health solutions in resource-constrained settings.

CCS Concepts

• Human-centered computing; • HCI design and evaluation methods; • Field studies;

Keywords

design methods, participatory design, digital health, self-determination theory, low-resource settings, mHealth design, community health workers

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1 Introduction

1.1 The challenge of dementia and co-morbidities in Latin America

Latin America is projected to experience the highest increase in dementia incidence globally, rising from 7.8 million in 2013 to 27 million by 2050 [1]. In Peru alone, approximately 300,000 people are living with dementia (PWD) [31]. These individuals typically have twice as many comorbidities as those without dementia. Despite being treatable, these comorbidities are often overlooked, leading to poor management and reduced quality of life for both PWD and their caregivers [2]. While there are currently no known cures for the illnesses that cause dementia, early diagnosis remains crucial for making essential decisions about treatment, care, support, and future planning.

Dementia is not a disease per se, but rather a state of psychosocial dysfunction and vulnerability. Its multidimensional nature affects various aspects of an individual's life and society at large, increasing both healthcare necessities and socioeconomic demands. This complexity compounds the challenges and inequalities prevalent in rural areas of low-resource settings, where health systems are often inadequate, lacking trained experts and essential diagnostic tools [3]. Peru, like many other low-resource countries, struggles to effectively support the diagnosis and treatment of PWD. Consequently, family and community members must assume more significant roles in care provision. It is in this context that appropriate mobile health (mHealth) technologies could offer much-needed support [32].

1.2 Socio-technical consideration for health systems design in low resource settings

While technological innovations like speech classifiers [11] and predictive models advance dementia diagnostics [12], purely technical approaches may have limited impact in resource-constrained settings [8]. A holistic, socio-technical, and participatory approach is crucial. Community and stakeholder engagement initiatives (CEI) [13], endorsed by organisations like National Institute for Health and Care Research (NIHR), enhance the relevance, acceptability, and sustainability of mHealth interventions. CEI provides a framework for addressing socio-technical considerations that naturally aligns with and recommends participatory design approaches. This approach ensures ethical and equitable implementation, aligning solutions with target population needs and balancing benefits and risks for vulnerable populations [9, 10, 14, 24].

Successful implementation of mHealth interventions is often hindered by methodological shortcomings, including lack of clear, replicable design processes and insufficient explanation of intervention components [13, 16, 22, 23]. Additionally, the absence of designers in the process often leads to inadequate application

of human-centered design principles, resulting in limited understanding of local contexts and stakeholder motivations. This is particularly crucial in diverse settings like Peru, with its 55 different indigenous ethnicities [17]. Research on co-designing health interventions for the Global South is scarce, leaving a gap in methodologies that balance community input with designers' decisions.

Moreover, mHealth interventions frequently overlook psychological needs such as autonomy, relatedness, and competence, which are essential for wellbeing and technology adoption [18, 33]. Barriers to user adoption persist, including usability issues, privacy concerns, and lack of digital literacy [26]. Implementation pose significant challenges, with many mHealth solutions struggling to integrate into existing health systems and workflows. Isolated pilot projects often fail to consider broader context and long-term impact, limiting scalability [21]. Addressing these limitations requires combining methodologies, deep cultural understanding, and strategic implementation planning for both implementation and scalability.

1.3 Research objective and overview

This case study explores a community-based, culturally aware approach to developing an mHealth intervention for dementia screening in Peru. We present an 18-month process structured using the Information Systems Research (ISR) framework [27, 28], a methodology for conducting and evaluating design science research in information systems, incorporating Human-Centered Design (HCD) methods.

Our study encompasses three phases: (1) pre-field studies, (2) field studies, and (3) post-field design and development. We describe the resulting mHealth tools, which are currently being implemented to screen 32,000 older adults across representative cities of the cultural diversity in Peru.

This paper focuses on the design process and its impact, aiming to share key learnings for researchers and practitioners working on similar projects in resource-constrained settings. Implementation and evaluation details are beyond the scope of this paper.

2 Background

This study is part of the IMPACT Salud project (Innovations using Mhealth for People with dementia and Co-morbidities) [4]. The project aims to strengthen health systems in Latin America using dementia as a tracer condition, through sustainable, integrated, person-centered, community-delivered, technology-enabled innovation. Our focus is on developing, testing, and implementing an mHealth-enabled system for dementia screening and diagnosis, while assessing its cost-effectiveness [5].

The project operates in four diverse regions of Peru: Lima, a megacity with over 10 million inhabitants; Tumbes, a semi-urban coastal region in northern Peru; Huancayo, an Andean city at 3200m altitude; and Iquitos, the main population center of the Peruvian jungle. These sites were chosen to represent diverse populations and leverage previous research experience in these regions.

3 Phase 1: Pre-field Studies

The pre-field phase, conducted from April to June 2023, focused on the ISR Rigor Cycles. ISR guides the iterative development and

refinement of technological solutions through cycles of relevance (contextual requirements), design (artifact creation and evaluation), and rigor (knowledge application and generation).

This phase included scoping, literature review, benchmarking, remote community meetings, low-fidelity prototyping, and field planning. Our primary goal was to develop a series of minimum viable products (MVPs) for field testing.

3.1 Setting and stakeholders

We engaged IMPACT Salud project researchers, healthcare practitioners, and external experts in digital healthcare and dementia care and professionals from Peru's Ministry of Health. Remote engagement with field site supervisors offered initial insights into local contexts.

3.2 Methodology

The pre-field phase consisted of a series of activities aiming to develop a set of MVPs that could serve as flexible working prototypes for the field work:

- *Semi-structured workshop (3 days, April)*. Conducted with the IMPACT Salud team to encourage divergent thinking and gather diverse perspectives. This enhanced internal collaboration and clarified design's role in global health and technology.
- *Literature review and benchmarking (April – June)*. Analysed existing mHealth tools in low-resource settings, identifying research gaps, design successes and failures. We created a structured visual, interaction, and user flows moodboard to inform UX/UI design.
- *Semi-structured interviews (15 meetings, 30 mins – 1h each, April – June)*. Held remote meetings with internal stakeholders to design user journeys, identify design frictions, formulate research questions, and gather suggestions for community participatory activities.
- *Expert interviews (10 interviews, 30 mins each, May – June)*. Conducted with external researchers, designers, and engineers from fields like digital healthcare, dementia care, and human-centered design in indigenous communities. These validated our process and informed user study structuring.
- *Remote engagement with site supervisors (8 meetings, 1h each, May – June)*. Gained deeper context understanding, established CHW connections, and collected insights for prototyping.
- *Wellbeing Supportive Design workshop (4h, June)*. Held a half-day intensive workshop to validate the MVP, focusing on enhancing human motivation and wellbeing for all stakeholders.
- *MVP Development (May – June)*. Prototyped the first MVP within a month using Airtable for data modelling and Glide for interface design. Adopted an iterative approach, preparing multiple MVP versions and design probes for field testing. Tested initial iterations with internal IMPACT stakeholders.

3.3 Outcomes

3.3.1 Research questions. What are the design requirements for an effective and accessible mHealth system for dementia screening for

low-resource settings? This question aimed to identify key design requirements, considering usability and engagement for CHWs and families. It aimed to explore features able to accommodate varying literacy and tech proficiency, while addressing cultural context and digital literacy in diverse Peruvian communities.

How might we optimise the mHealth tool to support and enhance CHWs' workflows? We examined CHWs' perceptions, motivations and concerns through MVP testing, investigating integration with existing processes and adoption barriers. The focus was on supporting CHWs' autonomy and efficiency in the field.

What technical and implementation challenges need to be addressed for successful deployment in remote areas? This explored technical constraints in remote areas, strategies for functionality with limited connectivity, and community relationships with technology.

These research questions shaped our field work methodology and informed our approach to MVP testing and iteration, ensuring we explored both technical aspects and cultural fit within the communities.

3.3.2 User journey. During the pre-field phase, we developed a user journey map to visualise the CHWs' workflow. This preliminary map served multiple purposes:

1. It provided a baseline understanding of existing CHW processes.
2. It highlighted potential gaps and opportunities in current practices.
3. It facilitated clear communication of our understanding to all project stakeholders.

The map was designed as a living document, intended to evolve throughout the entire process. While we anticipated significant changes during the field trip, this initial version helped us identify key touchpoints in the CHWs' workflow and areas where the mHealth tool could potentially streamline or enhance their activities.

3.3.3 Early pain points and opportunities. The pre-field phase revealed diverse challenges and opportunities across technical, clinical, cultural, and psychological domains. We identified issues in selecting appropriate screening questions, in addressing connectivity problems and adapting to varied weather conditions. Cultural considerations included building community trust, ensuring CHW safety in urban areas, and navigating language barriers. We noted non-standardised workflows among CHWs and varying levels of community relatedness. Privacy concerns about technologies like WhatsApp were raised. Opportunities emerged to support CHWs' psychological needs by enhancing autonomy, fostering competence, and strengthening community relatedness. These insights were crucial in shaping our field research approach and initial MVP design.

3.3.4 The minimum viable product. We rapidly prototyped the first MVP using Airtable (a cloud-based spreadsheet-database hybrid) for data modeling and Glide (a no-code app builder) for the interface. This no-code approach allowed us to create a functional product with core features, despite some UI limitations. Healthcare professionals and researchers from IMPACT Salud designed the

screening content, ensuring cognitive tests were accessible and culturally appropriate for Peru.

We then adopted an iterative approach, developing multiple MVP versions for field testing, accompanied by design probes to stimulate discussion. Initial iterations were tested with internal IMPACT stakeholders. This process yielded an MVP robust enough for initial testing yet flexible for real-world modifications.

4 Phase 2: Field studies

The field trip focused on the ISR Relevance and Design Cycles. We spent one week at each of four sites in Peru, structuring each visit into four key days: Discovery, Training and Co-ideation, Role Play, and Testing. We employed a mix of HCD and ethnographic methods throughout the process to ensure a comprehensive understanding of the local context and user needs and test our MVPs.

4.1 Participants and stakeholders

At each site, we engaged with 8-10 CHWs, 1-2 supervisors as gatekeepers, and 1 Medical Doctor. Participants were diverse in literacy levels, gender, and age. We collaborated with the IMPACT Salud Team, Instituto Peruano de Neurociencias representatives, and the Peruvian Ministry of Health's technology team. Throughout the fieldwork, we maintained ongoing communication with partners, providing regular progress updates and sharing initial findings.

4.2 Methodology

We employed a mixed-methods approach, aimed to identify problems, opportunities, and existing systems while testing MVPs and developing culturally sensitive iterations. Table 1 shows the methods used. The four-day cycle at each site included:

- *Day 1 - Discovery:* Rapid ethnography for context understanding.
- *Day 2 - Training and Co-Ideation:* Training sessions, knowledge sharing, journey mapping, and ideation exercises.
- *Day 3 - Role Play:* Service role play with MVPs.
- *Day 4 - Testing:* Field testing with MVPs to uncover undeclared opportunities and challenges.

4.3 Outcomes

4.3.1 Pain points, opportunities and design requirements. Key findings were categorised into themes such as offline functionality, usability and accessibility, user management, task management, data collection, user support, motivation and engagement, safety and context awareness, supervisor integration, workflow flexibility, internationalisation, family engagement, cultural adaptation, and dementia education support. For a detailed breakdown of the identified pain points, opportunities, and corresponding design requirements, please refer to Table 2 in the Appendix. This comprehensive table outlines the specific challenges encountered in each category and the potential design solutions developed to address them.

4.3.2 Minimum viable product iterations. The design team iteratively refined the MVP throughout the fieldwork, developing five versions using Glide. Using a web-app meant CHWs could easily run at-home training. The design team concurrently evolved the

Airtable data model, enabling quick structural adjustments. We also provided participants with paper concepts and Figma prototypes to explore interactions that went beyond Glide's capabilities.

Tangible prototypes were crucial for early identification of contextual issues and real-world usage observations. This approach maintained familiar workflows while enhancing efficiency and data quality [25]. At the conclusion of the field trip, results and prototypes were presented to the IMPACT Salud team, aligning key learnings and engaging stakeholders in identified necessary actions.

5 Phase 3: Post-field Design and Development

The eight-month post-field phase involved data analysis, design requirement compilation, feature backlog creation, and mHealth tool development. We maintained continuous community involvement, aligning with user needs and contextual realities. The interplay of Relevance, Design, and Rigor cycles guided our translation of field insights into design.

5.1 Participants and stakeholders

We sustained engagement with CHWs, supervisors and IMPACT Salud team throughout. Four months post-fieldwork, we began collaborating with outsourced developers, one month prior to official development.

5.2 Methodology

The post-field phase focused on design finalisation through iterative testing. We refined the MVP using field insights, thematic analysis, and wellbeing design guidelines [29]. Updated user journeys and design requirements were presented to stakeholders for input. After internal discussions, we engaged developers to assess feasibility and resources. We created a comprehensive features backlog, detailing app functionalities. We adopted a sprint-based approach for parallel design and development, designing system architecture and selecting Google Material as our design library. Jira, Slack, Click-up, and GitHub facilitated project management.

The research team provided crucial input on the screening flow and data collection, creating flow charts and a detailed dictionary for over 1000 variables collected during screening.

Throughout 15 design iterations, we shared remote updates with all the stakeholders, ensuring smooth transition from field insights to development while maintaining their involvement.

5.3 Outcomes

Our design process yielded a comprehensive socio-technical system for dementia screening, addressing the pain points and opportunities identified during fieldwork (see Table 2 in Appendix A.2). The system integrates an offline-first mobile app for CHWs, a web tool for supervisors, and a multi-tiered architecture.

5.3.1 System architecture. We designed a multi-tiered system architecture for the mHealth tool, prioritising robustness, scalability, and maintainability. The mobile app uses React Native for Android, while web components employ React (frontend) and Django (backend). PostgreSQL serves as the database on AWS RDS. The system leverages various AWS services, with Terraform and Ansible managing infrastructure. We integrated Metabase for data visualisation

Table 1: Field mix-methods

Method	Participants	Description and specific methods	Duration	Day	Purpose
(Rapid) Ethnography	CHWs, supervisors	Employed <i>participant observation</i> , <i>contextual interviews</i> , <i>five whys</i> , <i>in-depth interviews</i> , and <i>focus groups</i> to explore dementia stigma, system efficacy, literacy, motivation, technology challenges, and community dynamics. Local solutions were investigated and benchmarked.	2-4 hours	1, 2, 3	Gain deep understanding of users' context, behaviours, and needs
Journey Mapping	CHWs, supervisors, IMPACT	Collaboratively mapped CHWs' workflow across all screening stages. Served as a probe and reference tool for comprehensive process understanding.	30-60 mins	1, 2	Visualise CHWs' workflow, identify pain points and opportunities
Ideation based on analogies and association	CHWs, supervisors	Used random words, phrases, and images to trigger ideation on site-specific topics such as WhatsApp use, and successful volunteering experiences	60-90 mins	2, 3	Generate concepts, encourage creative thinking
Wizard of Oz	CHWs, supervisors	Simulated early-stage features like field planning and CHW messaging using <i>fast mock-ups</i> to test reactions.	60-90 mins	2, 3	Ideate and test potential functionalities
Service Role Play	CHWs, supervisors	CHWs enacted scenarios with the mHealth tool, using <i>experience prototyping</i> and <i>investigative rehearsal</i> to simulate real-world use.	60-90 mins	3	Test functionalities, usability, literacy, and paper vs. digital
Cognitive Walkthrough	CHWs, supervisors	CHWs performed specific tasks with the mHealth tool while researchers observed and noted usability issues.	45-60 mins	3, 4	Evaluate usability, identify navigation issues.
Piloting	CHWs	<i>Non-participant observation</i> of CHWs using the MVP in real settings, noting patterns, challenges, and adaptations.	2-3 hours	4	Field testing of MVP, identify undeclared challenges
Fast Prototyping	CHWs, supervisors, IMPACT	Rapid iteration of designs using sketches, paper prototypes, and digital mock-ups based on immediate user feedback.	30-60 mins	2, 3, 4	Quick iteration on ideas, validate concepts with users, and refine designs

and Google Maps for location services. This architecture supports offline functionality and incorporates security features. Cloudflare enhances security and performance, ensuring the tool meets current needs while allowing future adaptations.

5.3.2 mHealth App for Community Health Workers. The IMPACT Salud App, developed through iterative, human-centered design, is an offline-first data collection tool for CHWs in low-resourced Latin American settings. This Android tablet app facilitates door-to-door dementia screening. It integrates cognitive tests adapted by IMPACT Salud researchers. It supports CHWs in minimising unintended data collection errors, facilitating easy navigation of collected family data, managing and plan visits, navigating both

familiar and unfamiliar areas, and efficiently accessing and filtering collected family information.

The app offers comprehensive health visit management with on-line and offline functionality, ensuring data integrity in areas with limited connectivity. The app's main pages include recommended visits, all visits, map, calendar, settings, and backup, providing a simple interface for CHWs to manage their work efficiently. Questionnaire data collection pages are crafted to be accessible, accommodating varying literacy and tech levels. Questions navigation is facilitated through a multistep process and a highly recognisable summary page, which doubles as a data navigation tool.

Incorporating cultural nuances and local context, the app design enhances CHWs' sense of competence, autonomy, and relatedness

[33]. Personalisation features, such as the dynamic homepage or personalisable “i” tooltips, accommodate varying literacy levels and technological proficiency.

By streamlining workflows and addressing previously identified inefficiencies, the IMPACT Salud app aims to improve both efficiency and CHWs’ psychological well-being, recognising their crucial role in community health.

5.3.3 Web App for dementia diagnosis for Supervisors and Researchers. We developed a complementary web application for supervisors and researchers to manage and analyse collected data. This responsive app, accessible on various devices, offers role-based access for data privacy and relevance.

Field supervisors can access a dashboard displaying team performance metrics, visualise and edit data, and use an interactive map for strategic planning. Additional features include CHW family assignment and a dynamic FAQ section.

Researchers have expanded functionalities, including cross-site data access and advanced analysis tools. A key feature is the user-friendly questionnaire editor, allowing the IMPACT Salud team to modify or create surveys autonomously, potentially extending the system’s utility beyond dementia studies. By enabling the team to modify and expand the app independently, we foster a sense of ownership and encourage innovative applications beyond the initial scope, ensuring the tool’s long-term relevance and sustainability in the local context.

6 Lessons Learned

6.1 On the process

Continuous CHWs involvement is crucial for effective mHealth tool design: Involving CHWs throughout the design process was essential for ensuring the mHealth tool met their needs, aligned with existing workflows, and improved their confidence and motivation. Iterative prototyping proved invaluable for stimulating discussion and refining features. A cultural and linguistic gatekeeper proved essential for community access and activity translation.

Acknowledging participant input while supporting project requirements: Participatory design processes often reveal a tension between individual CHW suggestions and overarching project goals. While CHWs provide invaluable context-specific insights, their proposed solutions may not always align with broader project requirements or established HCI principles. CHWs initially requested a single, long page listing all families. Instead, we implemented a search and filter system, balancing their need for comprehensive access with more efficient information retrieval [37]. To effectively balance CHWs’ input with design requirements, we: documented CHWs’ suggestions, evaluated them in context, and transparently communicated how their input shaped final decisions and ensuring CHWs understood the rationale behind design choices and felt ownership of the solutions.

Training local facilitators enhances insight gathering: Techniques like echoing and boomerang [31] significantly improved workshop outcomes.

Cultural sensitivity in language aids destigmatisation: Addressing mental health stigma requires careful consideration throughout the field work. Simple changes in terminology, like referring to

“families” instead of “patients” or “carers,” aided destigmatisation efforts and improved engagement.

Flexible planning is essential for field study in diverse settings: Field conditions varied significantly due to weather, cultural barriers, and varying levels of education and digital literacy. Heat and rain affected both technology performance and participant engagement. We recommend planning for double the anticipated time and preparing for interruptions in the field studies. Additionally, scheduling a week of rest and prototyping between site visits proved essential, as the intensity of consecutive field studies was underestimated [36].

Visual and tangible materials are crucial in the participatory process [35]: Materials like prototypes and probes facilitated effective communication and ideation when faced with time constraints and potential cultural barriers.

Understanding CHW motivations is crucial for successful engagement throughout the process: CHW motivations varied, influencing both feature design and workshop engagement. Motivation fluctuated based on local recognition, personal experience, competing responsibilities, and family relationships [34]. Providing for basic needs during training and field work (e.g., sunscreen, hats, water, insect repellent) also proved important for motivation. Factors like lack of training, technical support, and poor internet connectivity negatively impacted motivation. Offering choices (e.g., between tablets and phones) increased CHWs’ sense of autonomy.

Consistent communication is vital in multidisciplinary teams and challenging environments: While a multidisciplinary team is crucial, aligning diverse perspectives presented challenges. Emphasising the iterative nature of the process and managing expectations helped maintain team cohesion throughout the project.

6.2 On the products

Start with MVPs and iterate rapidly to refine the product: Beginning with simpler versions helped validate concepts, gather feedback, and refine the tool effectively while minimising resource waste [38].

Essential technical features for low-resource settings must be considered from the start: Key technical requirements included: offline functionality, adhere to high usability and accessibility standards, flexibility to adapt to varying literacy levels and technological proficiency and scalability. These aspects need to be taken into consideration from the project’s inception to ensure the tool’s effectiveness in diverse environments.

Prioritising scalability and long-term sustainability may increase initial costs but yields future benefits: Developing a scalable app with features like a questionnaire editor was more complex but offered long-term benefits [39], including potential ownership by our stakeholders.

Field testing and ongoing support are crucial for successful mHealth implementation: To ensure adoption, it’s essential to allocate budget for maintenance and conduct field tests as early as possible with real users in actual conditions, including varying internet connectivity and online/offline scenarios. Continuous training and support during initial mHealth deployments are crucial for CHWs to become comfortable and competent with the tools.

Utilising standard design libraries enhances accessibility and scalability: In terms of user interface design, using well-known and widely used libraries with standard accessibility measures, such as Google Material, improved both accessibility and scalability from a design perspective. Visual aids, iconography, and progress tracking features proved beneficial for intuitive navigation and user engagement.

Integrating training elements into the interface might be more effective than only separate sessions: In our case, it was more effective than relying solely on separate training sessions. This approach provided users with immediate, contextual guidance.

Design must reflect real community experiences: The mHealth tool's design needed to account for real community experiences and challenging weather conditions in low-resource settings. Designers and researchers should support the transition from previous work methods to the new digital tools.

Comprehensive training is as crucial as the product itself: Training emerged as a key factor in the success of the mHealth tool. It should cover not only the app's functionality but also topics like dementia awareness, appropriate interaction with families and address potential cultural adaptations made by CHWs that could affect results, ensuring consistency while respecting local contexts.

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A Appendices

A.1 Useful Link

For more detailed information about the IMPACT Salud project and to explore our mHealth tool, please refer to the following resources:

- Impact Website: <https://impact-salud.org>
- Impact Salud Video Project Presentation: <https://vimeo.com/953583601>
- mHealth hi-fi Prototype: <https://shorturl.at/Ct5hj>

A.2 Pain points, opportunities and design requirements – Table 2

Table 2: IMPACT Salud App Design requirements

Requirement Category	Pain points, opportunities	Design requirements, potential features
Offline Functionality	Unreliable internet connection Data loss concerns	Offline-first design Automatic data saving Sync status indicators
Usability & Accessibility	Varying literacy levels Tech literacy challenges UI issues (scrolling, text size), Eyesight problems Need for clear test results display	Clear, icon-based interface Read-aloud function for questions Simplified interface with essential elements App walkthrough and embedded training Adjustable text size and spacing Voice recording and transcription
User Management	Need for quick access maintaining data safety Privacy concerns	Biometric login (Face ID, fingerprint) User profiles with customisable information
Task Management	CHWs need to be autonomous on the field Difficulty managing patient lists Need for visit planning and tracking Forgetting time-sensitive questions	Summary page Dashboard with visit suggestions Map view of visits Homepage suggested visits Filters and search for visits Patient information editing in overview area Calendar integration Pop-up reminders for time-sensitive questions
Data Collection	Paper-based note-taking habits Need for flexible data entry Influence of family members on responses Confusion about certain questions	In-app note-taking feature Complete later function Standardised training to reduce bias Discover more button
User Support	Need for ongoing training Struggle with consent form understanding Struggle when the app does not work	In-app help sections, FAQs Example provided for confusing questions Video training sessions Feedback mechanism to supervisors Simplified consent forms with clear explanations Notifications on bugs solving status
Motivation and Engagement	Need for recognition Desire to track progress Importance of feedback loop	Certificates/proof of work/welcome page Performance statistics Progress tracking within questionnaires Regular feedback sessions with supervisors Feedback loop channels for help
Safety and Context Awareness	Safety concerns in certain areas Need for context awareness when not known Fear of damaging equipment	Map showing other CHWs and health centres Two-CHW for dangerous areas Ability to pause/resume tasks Tablet hidden in pockets Robust tablet covers
Supervisor Integration (for supervisors and researchers)	Need for supervisor oversight Research team working in silos	Dashboard for supervisors and researchers Ability to assign and track CHW tasks Updated socio-technical system map for stakeholders
Workflow Flexibility	Different management styles Need for visit status tracking	Customisable workflow options Multiple visit status options (e.g., pending, completed, partially completed, planned, no response, stopped)
Internationalisation	Need for use in different regions	Multi-language support Culturally adaptable interface elements

Family Engagement	Need for family understanding of consent and aim of the screening Varying levels of health literacy among families	Educational modules for families Simplified visual explanations of dementia and screening process Option for family members to provide feedback on the screening experience
Cultural Adaptation	Diverse cultural contexts across Peru Need for community-specific approaches	Customisable interface with local imagery Community co-design workshops for region-specific adaptations Tailored examples and training
Dementia Education Support	CHWs struggle to explain dementia to families Need for culturally appropriate explanations	Illustrated in-app guides for explaining dementia simply Cultural-specific analogies and examples library Short, shareable videos explaining dementia

A.3 ISR Framework

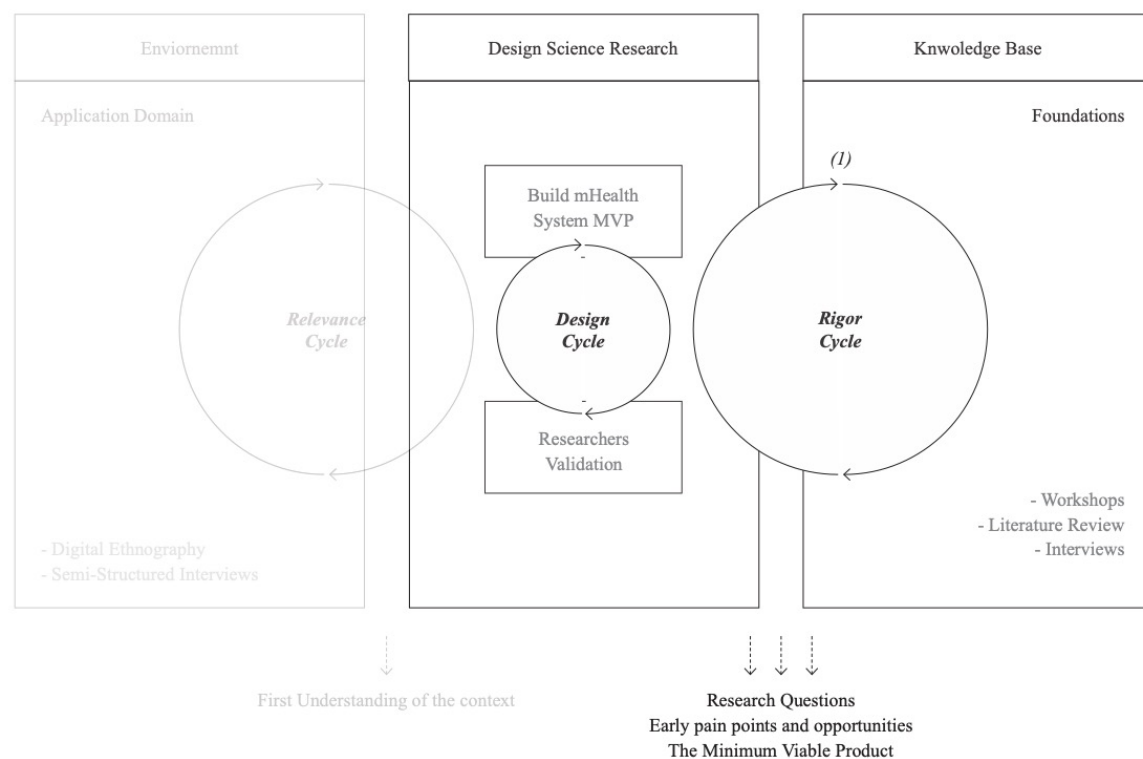


Figure 1: ISR Diagram (Design Research Cycles) in the Pre-field

Pre-field ISR Diagram (1): This diagram illustrates the initial phase of the ISR process, focusing on foundation-building and preliminary research. The Rigor Cycle encompasses workshops, literature reviews, and interviews, feeding into the Design Cycle where an mHealth System MVP is built. The Relevance Cycle involves digital ethnography and semi-structured interviews. These cycles converge to generate the first understanding of the context, research questions, early pain points and opportunities, and the MVP.

Field ISR Diagram (2): This diagram represents the field phase of the ISR process, emphasising hands-on research and rapid prototyping. The Relevance Cycle includes rapid ethnography, journey mapping, ideation, and Wizard of Oz techniques. The Design Cycle features service role play, cognitive walkthroughs, fast prototyping, and piloting. These activities lead to the identification of pain points, opportunities, design requirements, and MVP iterations, grounding the research in the application domain and real-world environment.

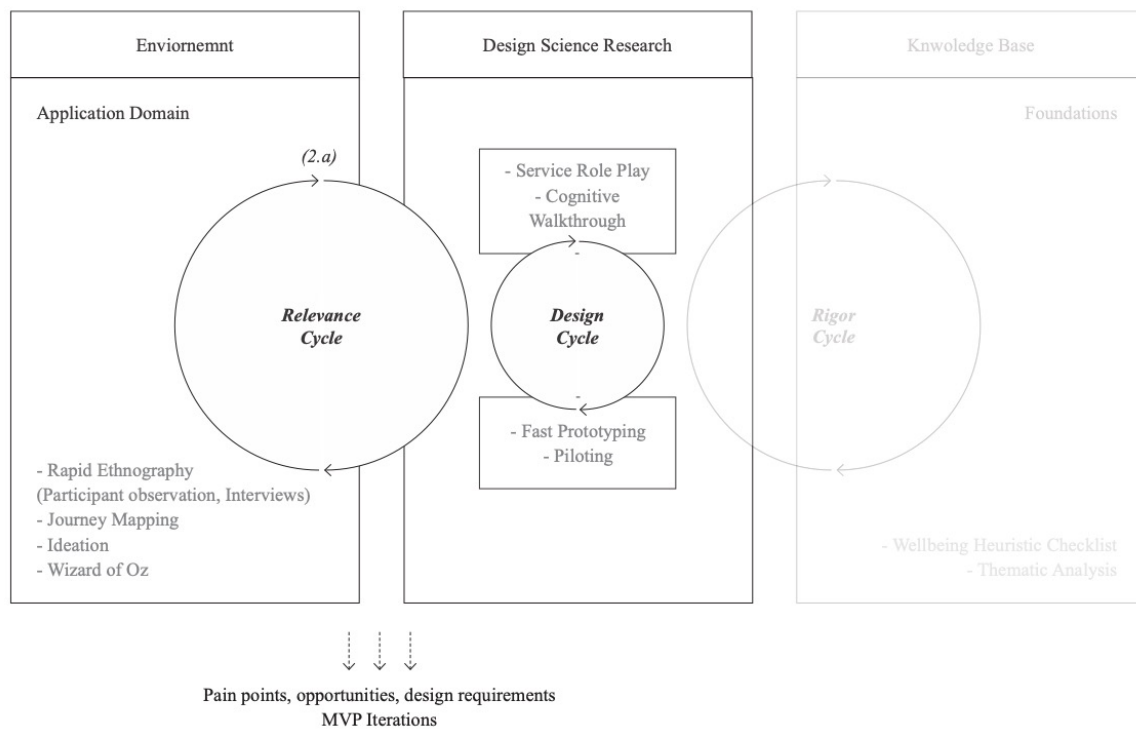


Figure 2: ISR Diagram (Design Research Cycles) in the Field

A.4 Pictures



Figure 3: Researchers using wellbeing cards to incorporate psychological needs into mHealth design



Figure 4: Design team engaged in rapid ideation using the Crazy 8 technique to generate diverse app concepts.

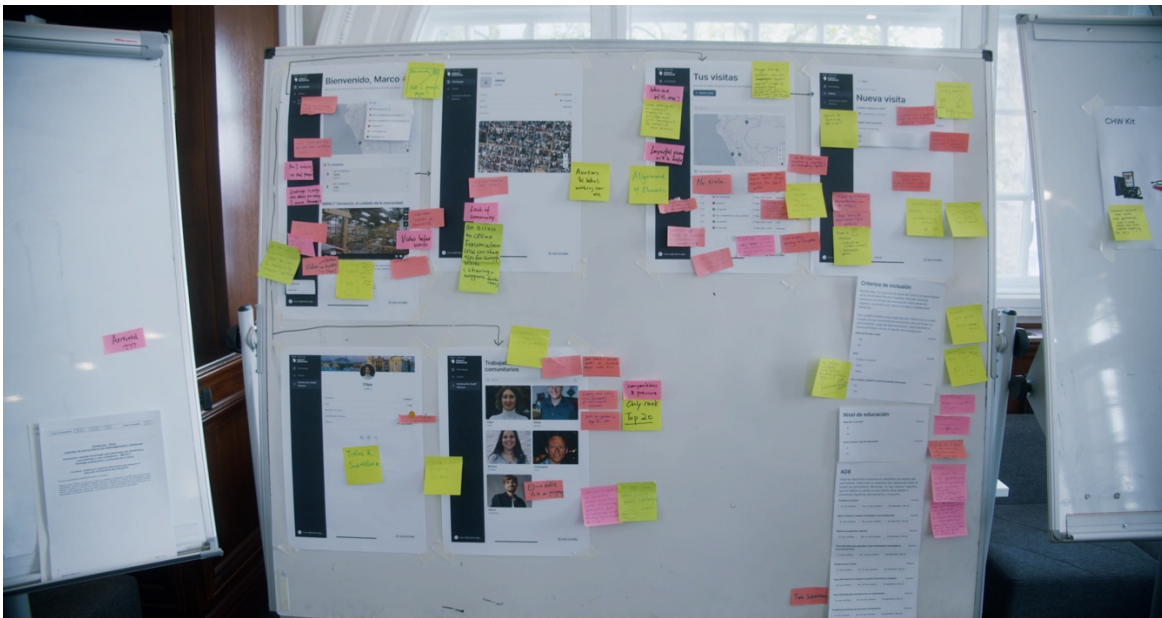


Figure 5: Visual mapping of user pain points and design opportunities identified during initial MVP testing.

A.5 IMPACT Salud App and Web App



Figure 6: IMPACT Salud members reviewing MVP prototype in Lima.



Figure 7: Nauta, near Iquitos, in the Peruvian Amazon.

A.5 IMPACT Salud App Prototypes



Figure 8: CHW training session in Villa El Salvador, Lima, introducing the cognitive tests.



Figure 9: Participatory service role play in Nauta, simulating real-world use of the mHealth app.



Figure 10: CHWs gathered outside a health center in San Jacinto, Tumbes.



Figure 11: CHW testing MVP in real-world conditions.



Figure 12: CHWs engaging with families in San Jacinto, Tumbes.



Figure 13: CHWs engaging with families outdoor, in San Jacinto, Tumbes.



Figure 14: CHW interacting with families in Iquitos.

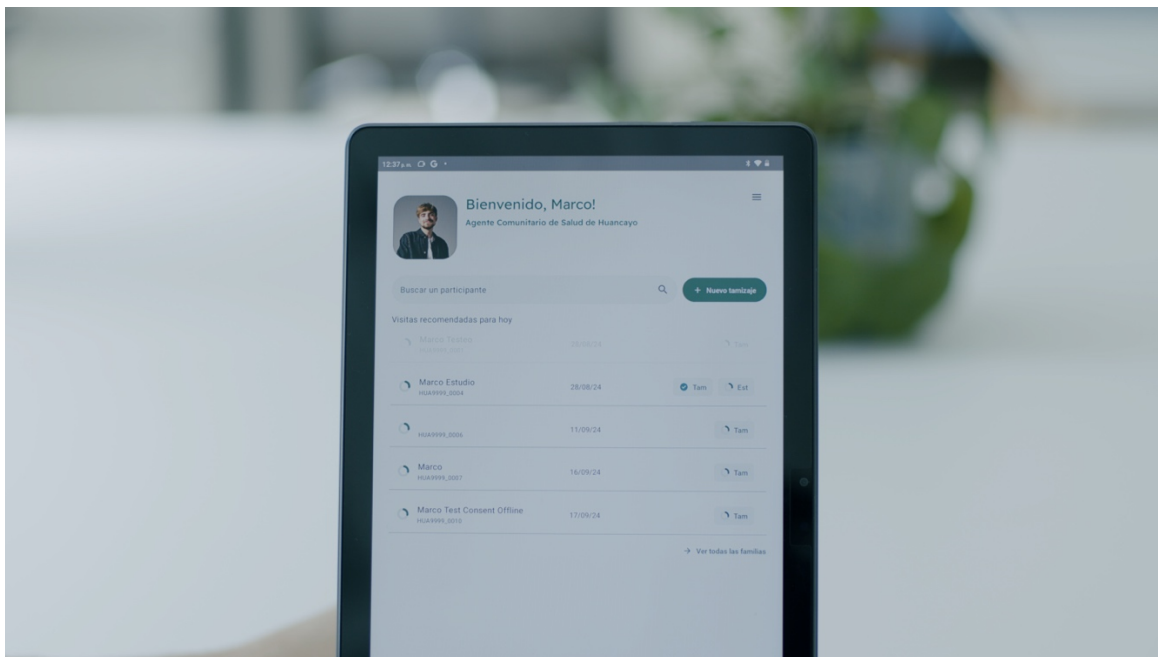


Figure 15: IMPACT Salud App, Welcome Screen, Standard Size.



Figure 16: Initiating new dementia screenings.

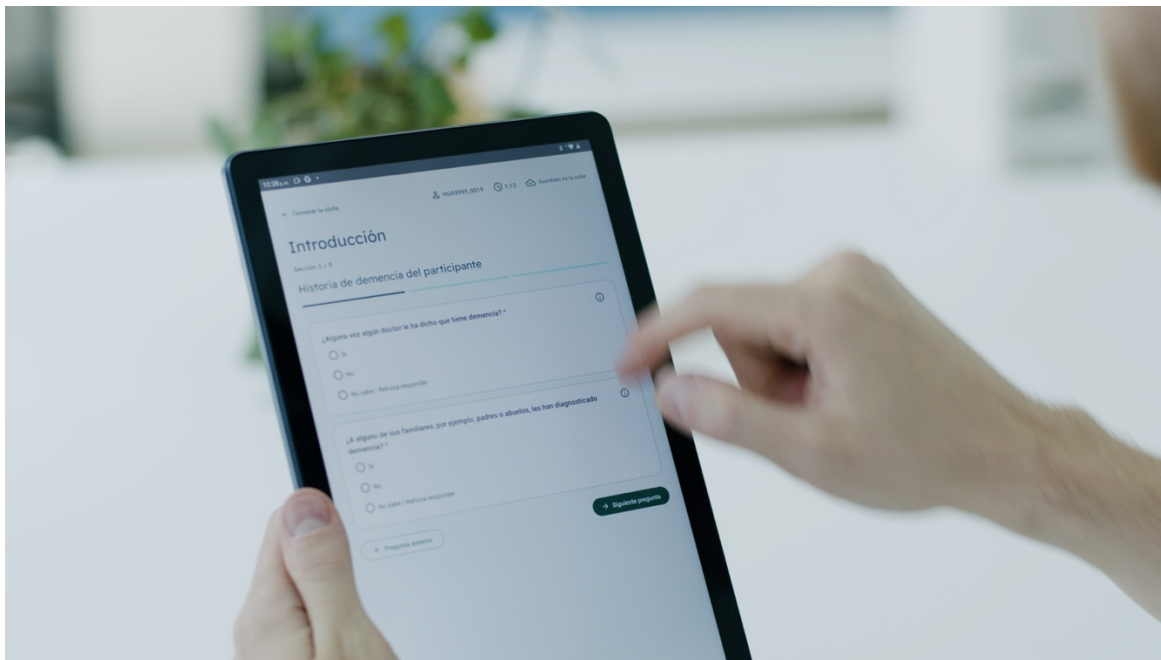


Figure 17: In-progress screening view.

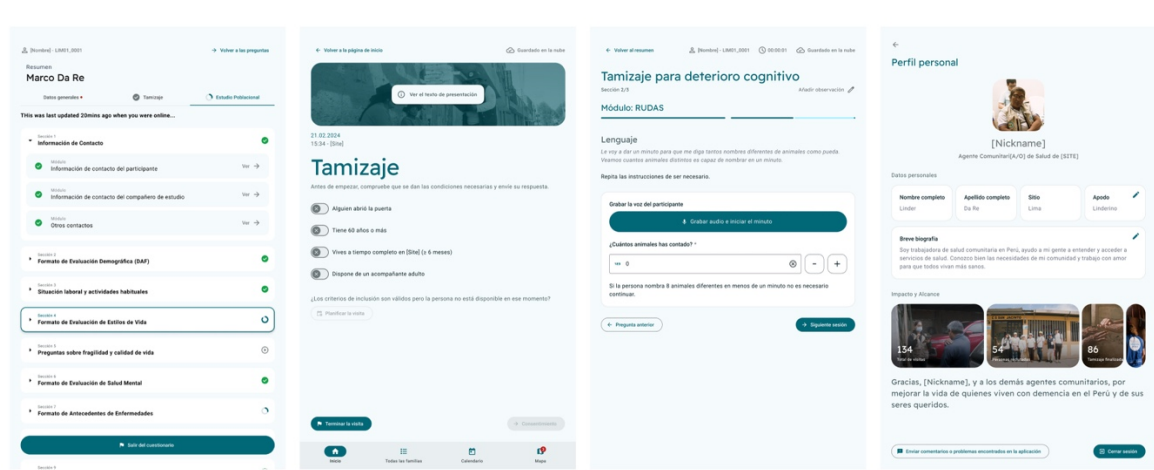


Figure 18: Some of the app features: (a) Summary page for quick overview, (b) Inclusion criteria checklist, (c) Voice recording and analysis for cognitive assessment, (d) Customisable personal profile page.

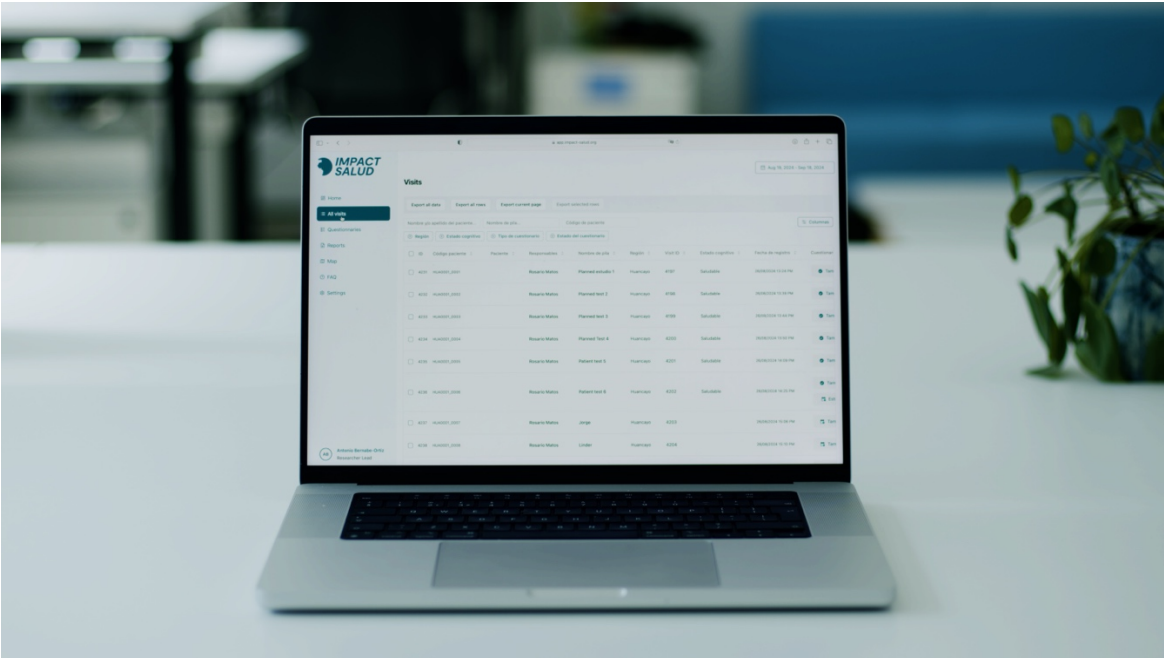


Figure 19: Web interface displaying organised family list, facilitating efficient data management for supervisors.

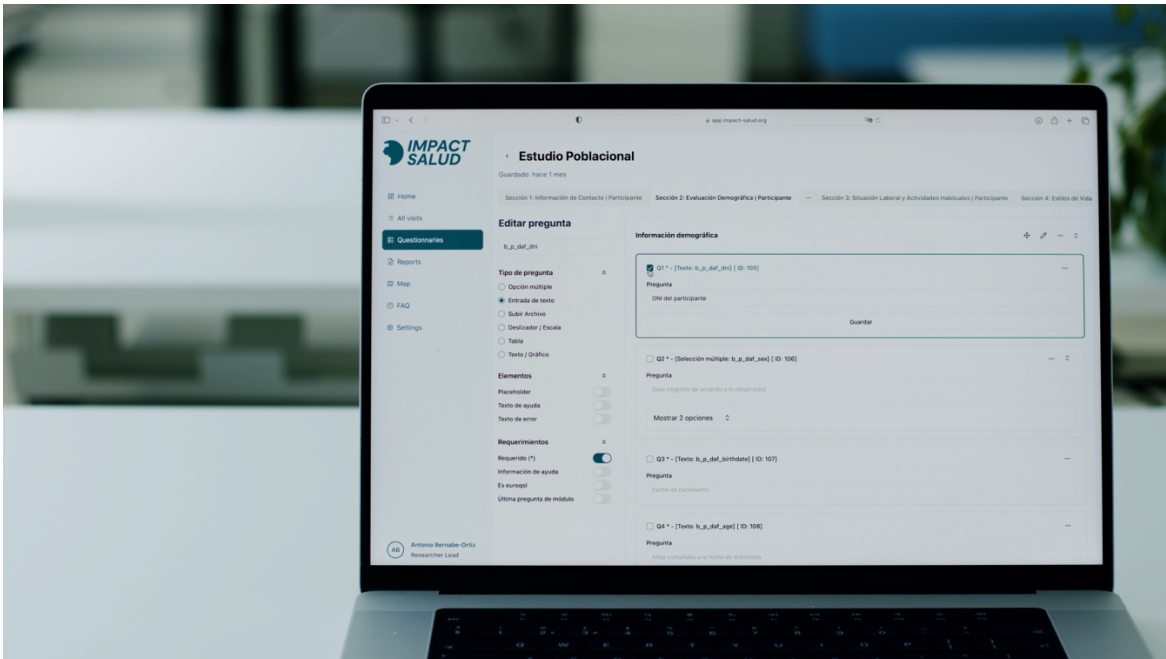


Figure 20: Web interface displaying the questionnaire editor with which researchers can autonomously edit the content of the app

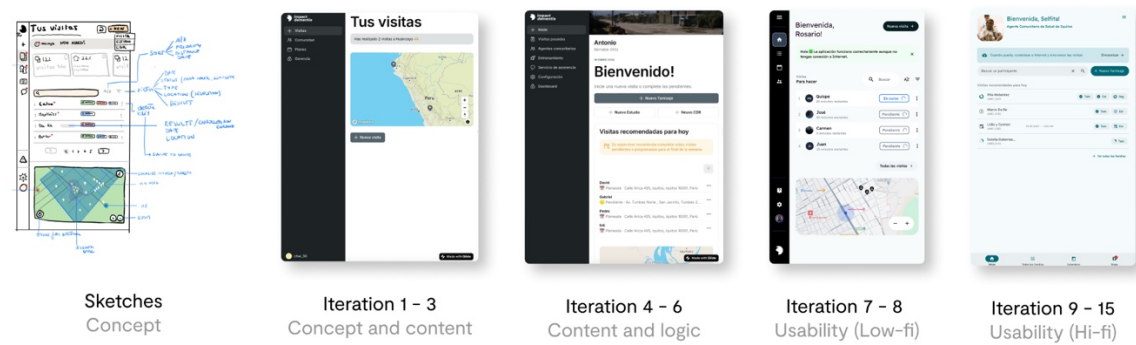


Figure 21: Impact SALUD App Prototypes.

Link to the paper interactive prototype: <https://shorturl.at/GhCfc>

Link to the MVP v2: <https://impact-map-2.glide.page>

Link to the MVP v4: <https://impact-mvp-3.glide.page>

Link to the MVP v6: <https://impactperu.glide.page>

Link to the interactive prototype v8: <https://shorturl.at/dxKdd>

Link to the interactive prototype v15: <https://shorturl.at/Ct5hj>